MODULE 2…

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QUES 1 :

```
FraudSystemDetection.java > FraudSystemDetection
import java.util.Scanner;

public class FraudSystemDetection {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] transaction = new int[n];

        for (int i = 0; i < n; i++) {
            transaction[i] = sc.nextInt();
        }

        int search = sc.nextInt();

        boolean found = false;

        for (int i = 0; i < n; i++) {
            if (transaction[i] == search) {
                found = true;
                break;
            }
        }

        if (found)
            System.out.println(x: "Fraud Transaction Found");
        else
            System.out.println(x: "Transaction Not Found");

        sc.close();
    }
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:
tection }
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 2 :

ServerLogSearch.java > ...

```java
import java.util.Scanner;

public class ServerLogSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] logs = new int[n];

        for (int i = 0; i < n; i++)
            logs[i] = sc.nextInt();

        int target = sc.nextInt();

        int low = 0, high = n - 1;

        while (low <= high) {
            int mid = (low + high) / 2;

            if (logs[mid] == target) {
                System.out.println("Log Found");
                return;
            }

            if (target < logs[mid])
                high = mid - 1;
            else
                low = mid + 1;
        }

        System.out.println("Log Not Found");
        sc.close();
    }
}
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
```

QUES 3 :

EmployeeSalarySort.java ×

EmployeeSalarySort.java > EmployeeSalarySort

```java
import java.util.Scanner;

public class EmployeeSalarySort {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] salary = new int[n];

        for (int i = 0; i < n; i++)
            salary[i] = sc.nextInt();

        // Bubble Sort
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (salary[j] > salary[j + 1]) {
                    int temp = salary[j];
                    salary[j] = salary[j + 1];
                    salary[j + 1] = temp;
                }
            }
        }

        // Print sorted salaries
        for (int i = 0; i < n; i++)
            System.out.print(salary[i] + " ");

        sc.close();
    }
}
```

OUTPUT :

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:\Users\Siddh
ort }
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 4 :

```
EmployeeSalarySort.java    WebsiteResponseSort.java ×
WebsiteResponseSort.java > WebsiteResponseSort
import java.util.Scanner;

public class WebsiteResponseSort {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] response = new int[n];

        for (int i = 0; i < n; i++)
            response[i] = sc.nextInt();

        // Insertion Sort
        for (int i = 1; i < n; i++) {
            int key = response[i];
            int j = i - 1;

            while (j >= 0 && response[j] > key) {
                response[j + 1] = response[j];
                j--;
            }

            response[j + 1] = key;
        }

        // Print sorted response times
        for (int i = 0; i < n; i++)
            System.out.print(response[i] + " ");

        sc.close();
    }
}
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c
eSort }
5
250 120 320 180 100
100 120 180 250 320
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 5 :

EmployeeSalarySort.java | WebsiteResponseSort.java | BugReportPriority.java

BugReportPriority.java > BugReportPriority

```java
import java.util.Scanner;

public class BugReportPriority {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] priority = new int[n];

        for (int i = 0; i < n; i++)
            priority[i] = sc.nextInt();

        // Selection Sort
        for (int i = 0; i < n - 1; i++) {
            int min = i;

            for (int j = i + 1; j < n; j++) {
                if (priority[j] < priority[min])
                    min = j;
            }

            int temp = priority[i];
            priority[i] = priority[min];
            priority[min] = temp;
        }

        // Print sorted priorities
        for (int i = 0; i < n; i++)
            System.out.print(priority[i] + " ");

        sc.close();
    }
}
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> co
y }
5
9 2 8 1 5
1 2 5 8 9
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

Date ___/___/___

Day________

Module 2

Q1)

```
import java.util.Scanner;
Public Class FraudSystemDetection {
    Public Static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt ();
        int [] transaction = new int [n];
        for (int i = 0; i<n; i++) {
            transaction [i] = sc.nextInt ();
        }

        int Search = sc.nextInt ();
        boolean Found = false;
        for (int i =0; i<n ; i++) {
            if (transaction [i] == Search ) {
                found = true;
            } break;
        }
```

Date __/__/__
Day ______

```
if (found) {
    system.out.println(x: "Fraud Transaction found");
}
else
    System.out.println(x: "Transaction not found");
Sc.close();
}
```

OUTPUT

→ 5

→ 1201 1305 1480 1408 1502

→ 1450.

Fraud Transaction found.

Date __/__/__

Day ______

Q2)

import java.util.Scanner;

public Class ServerlogSearch {

public static void main (String [] args){

Scanner sc = new Scanner(system.in);

int[] ~~int n = 5~~ logs = new int[n];

for (int i =0; i<n; i++)

log [i] = sc.nextInt();

int target = sc.nextInt();

int low = 0, high = n-1;

while (low <= high) {

int mid = (low + high)/2;

if (logs[mid] == target) {

System.out.println(x: "log Found");

return;

}

Date ___/___/___

Day________

```
if (target < logs[mid])
        high = mid - 1;
else
        low = mid + 1;
    }
System.out.println(x: "Log not found");
sc.close();
  }
}
```

OUTPUT

⇒ 7

⇒ 1001 1005 1010 1015 1020 1030 1040

⇒ 1015

⋙ Log Found.

Date ___/___/___

Day ________

Q3.)

```
import java.util.Scanner;
   public class EmployeeSalarySort {
      public static void main (String [] args) {
      Scanner sc = new Scanner (System.in);
      int n = sc.nextInt ();
      int [] = salary = new int [n];
      for (int i = 0; i < n; i++)
          salary [i] = sc.nextInt ();
      for (int i = 0; i < n-1; i++) {
         for ( int j = 0; j < n-i-1 ; j++) {
             if (salary [j] > salary [j+1]) {
                int temp = salary [j];
               salary [j] = salary [j+1];
               salary [j+1] = temp;
               }
         }  }
      }
```

Date __/__/__
Day____

```
for (int i = 0; i < n; i++)
    System.out.print(salary[i] + " ");
sc.close();
}
```

OUTPUT

⇒ 5

⇒ 45000 32000 55000 28000 60000

⇒ 28000 32000 45000 55000 60000

Date ___/___/___
Day ________

Q4.)

```
import java.util.Scanner;
public class websiteResponseSort {
    public static void main(String []args){
        Scanner sc = new Scanner(System.in);

        int n= sc.nextInt();
        int [] response = new int[n];

        for (int i=0; i<n; i++)
            response [i] =sc.nextInt();

        for (int i=1; i<n; i++){
            int key = response [i];
            int j = i-1;

            while (j>=0 && response[j]>key){
                response[j+1] = response [j];
                j--;
            }
```

```
response[j+1] = key;
}

for (int i=0; i<n; i++)
    System.out.print(response[i]+" ");
sc.close();
}
}
```

OUTPUT

→ 5

→ 250 120 320 180 100

→→ 100 120 180 250 320

Date ___/___/___
Day ________

Q5)

```
import
 java.util.Scanner;
public class BugReportPriority {
 public static void main (String[] args) {
  Scanner sc = new Scanner (system.in);
  int n = sc.neatInt ();
  int [] priority = new int [n];

  for (int i=0 ; i<n; i++)
      priority [i] = sc.neatInt ();
  for (int i = 0 ; i <n -1 ; i++) {
     int min = i;

     for (int j = i+1; j <n; j++) {
       if (priority [j] < priority[min])
         min = j;
     }
```

⟶

Date___/___/___
Day________

```
        int temp = priority [i];
        priority [i] = priority [min];
        Priority [min] = temp;
    }

for (int i = 0; i <n; i++)
    system.out.print(priority [i] + " ");

sc. close ();

  }
}
```

OUTPUT

⇒ 5

⇒ 9 2 8 1 5

>>> 1 2 5 8 9